**Department of Mechanical and Production Engineering**

Aarhus University

# Formula Sheet



| | |
|---|---|
| **Author:** | Noah Rahbek Bigum Hansen |
| **Student ID:** | 202405538 |
| **Course:** | Electromagnetism & Electronics — 290231U007 |
| **Date:** | February 11, 2026 |

# Contents


**1 Constant Definitions** **3**
1.1 Coulomb Constant . . . . . 3

**2 Lecture 1: Electrostatics** **3**
2.1 Coulomb's Law . . . . . 3
2.1.1 Principle of Superposition . . . . . 3
2.2 Quantization of Charge . . . . . 3
2.3 Electric Field at a point . . . . . 4
2.3.1 Electric Field Produced by Point Charge . . . . . 4
2.3.2 Electric Field due to a Dipole along dipole axis . . . . . 4
2.3.3 Electric Field due to a Dipole along Dipole axis at Long Distance . . . . . 4
2.3.4 Electric Field of a Charged Ring at Large Distance . . . . . 5
2.3.5 Electric Field of a Charged Disk . . . . . 5
2.3.6 Electric field from an infinite sheet . . . . . 5
2.3.7 Electric field of a Conducting Sphere . . . . . 6
2.4 Electric Field of an Insulating Sphere . . . . . 6
2.5 Gauss' Law . . . . . 7
2.5.1 Net electric flux through a surface caused by an electric field . . . . . 7
2.6 Electric Potential Energy . . . . . 7
2.6.1 Electrostatic Potential . . . . . 8
2.6.2 Electrostatic Potential of a Plate Capacitor . . . . . 8

**3 Lecture 2: Fundamentals of Electric Circuits** **8**
3.1 Current, Voltage, and Resistance . . . . . 8
3.1.1 Electric Current . . . . . 8
3.1.2 Electric Voltage . . . . . 8
3.1.3 Electric Resistance . . . . . 9
3.1.4 Current Density . . . . . 9
3.1.5 Resistivity . . . . . 9
3.1.6 Conductivity . . . . . 9
3.1.7 Resistance From Resistivity . . . . . 9
3.1.8 Ohm's Law . . . . . 10
3.1.9 Electric Power . . . . . 10
3.2 Kirchoff's Circuit Laws . . . . . 10
3.2.1 Kirchoff's Current Law (KCL) . . . . . 10
3.2.2 Kirchoff's Voltage Law (KVL) . . . . . 10
3.2.3 Resistors in Series . . . . . 10
3.2.4 Voltage Divider Rule . . . . . 11
3.2.5 Resistors in Parallel . . . . . 11
3.2.6 Current Divider Rule . . . . . 11

**4 Lecture 3: Principles of Circuit Analysis** **11**

**5 Lecture 4: Capacitors & Inductors** **11**
5.1 Capacitors . . . . . 11
5.1.1 Capacitance . . . . . 11
5.1.2 Capacitance of a Parallel-Plate Capacitor . . . . . 11
5.1.3 I-V Relation for an Ideal Capacitor . . . . . 12

5.1.4 Capacitance of a Cylindrical Capacitor . . . 12
5.1.5 Capacitance of a Cylindrical Capacitor . . . 12
5.1.6 Parallel Capacitance . . . 12
5.1.7 Capacitors in Series . . . 13
5.1.8 Potential Energy in a Capacitor . . . 13
5.2 Inductors . . . 13
5.2.1 Inductance . . . 13
5.2.2 I-V Relation for an Ideal Inductor . . . 13
5.2.3 Magnetic Field of an Inductor . . . 13
5.2.4 Inductors in Series . . . 14
5.2.5 Parallel Inductors . . . 14
5.2.6 Energy Stored in an Inductor . . . 14
5.2.7 Energy Density in an Inductor . . . 14
5.3 RC and RL Circuits in the DC Regime . . . 14
5.3.1 Charge in a (Charging) Capacitor as a Function of Time . . . 14
5.3.2 Current in a (Charging) Capacitor as a Function of Time . . . 15
5.3.3 Capacitive Time Constant . . . 15
5.3.4 Inductive Time Constant . . . 15
5.3.5 Current in an Inductor . . . 15

**6 Lecture 5: AC Circuits 16**
6.1 AC Circuits . . . 16
6.1.1 Periodic Signals . . . 16
6.1.2 Average Value of a Signal Waveform . . . 16
6.1.3 RMS value . . . 16
6.2 Impedance . . . 16
6.2.1 Impedance of a Resistor . . . 16
6.2.2 Impedance of an Inductor . . . 17
6.2.3 Impedance of a Capacitor . . . 17
6.2.4 Total Impedance . . . 17
6.2.5 Admittance . . . 17
6.3 Transient Analysis . . . 18
6.3.1 1st Order Differential Equation for RC Circuits . . . 18
6.3.2 2nd Order Differential Equation for RLC Circuits . . . 18

# 1 Constant Definitions

## 1.1 Coulomb Constant

$$k = \frac{1}{4\pi\epsilon_0} = 8{,}99 \cdot 10^9\,\mathrm{N\,m^2/C^2}.$$

### Symbols

| | | |
|---|---|---|
| $k$ | Coulomb constant | $8{,}99 \cdot 10^9\,\mathrm{N\,m^2/C^2}$ |
| $\epsilon_0$ | Permittivity constant | $8{,}85 \cdot 10^{-12}\,\mathrm{C^2/N\,m^2}$ |

## Lecture 1: Electrostatics

26. Januar 2026

## 2.1 Coulomb's Law

$$\boldsymbol{F} = k\frac{q_1 q_2}{r^2}\,\hat{r}.$$

### Assumptions

- Point charges

### Symbols

| | | |
|---|---|---|
| $\boldsymbol{F}$ | Electrostatic force | N |
| $k$ | Coulomb constant | $8{,}99 \cdot 10^9\,\mathrm{N\,m^2/C^2}$ |
| $q_1, q_2$ | Charges | C |
| $r$ | Distance | m |
| $\hat{r}$ | Unit vector | — |

### 2.1.1 Principle of Superposition

$$\boldsymbol{F}_{1,\mathrm{net}} = \sum_{i=1}^{n} \boldsymbol{F}_{1i}.$$

### Symbols

| | | |
|---|---|---|
| $\boldsymbol{F}_{1,\mathrm{net}}$ | Net force | N |
| $\boldsymbol{F}_{1i}$ | Force contribution on particle 1 from particle $i$ | N |

## 2.2 Quantization of Charge

$$q = \pm n \cdot e.$$

### Symbols

| | | |
|---|---|---|
| $q$ | Charge | C |
| $n$ | Integer number ($n \in \mathbb{Z}$) | — |
| $e$ | Elementary charge | $1{,}6 \cdot 10^{-19}\,\mathrm{C}$ |

## 2.3 Electric Field at a point

$$\boldsymbol{E}(x,y,z) = \frac{\boldsymbol{F}(x,y,z)}{q_0}.$$

### Symbols

| | | |
|---|---|---|
| $\boldsymbol{E}(x,y,z)$ | Electric field at point $(x,y,z)$ | $\frac{\mathrm{N}}{\mathrm{C}}$ |
| $\boldsymbol{F}(x,y,z)$ | Force on test particle at point $(x,y,z)$ | N |
| $q_0$ | Test charge | C |

### 2.3.1 Electric Field Produced by Point Charge

$$\boldsymbol{E} = \frac{1}{4\pi\epsilon_0}\frac{q}{r^2}\hat{r}$$
$$E = \frac{1}{4\pi\epsilon_0}\frac{|q|}{r^2}.$$

### Symbols

| | | |
|---|---|---|
| $E$ | Electric field strength | $\frac{\mathrm{N}}{\mathrm{C}}$ |
| $\epsilon_0$ | Permittivity constant | $8{,}85 \cdot 10^{-12}\,\mathrm{C}^2/\mathrm{N}\,\mathrm{m}^2$ |
| $q$ | Charge | C |
| $r$ | Distance | m |
| $\hat{r}$ | Unit vector | — |

### 2.3.2 Electric Field due to a Dipole along dipole axis

$$E(z) = E_{(+)}(z) + E_{(-)}(z) = \frac{q}{4\pi\epsilon_0 z^2}\left(\frac{1}{1-\frac{d^2}{2z}} - \frac{1}{1+\frac{d^2}{2z}}\right) = \frac{q}{2\pi\epsilon_0 z^3}\left(\frac{d}{\left(1-\left(\frac{d}{2z}\right)^2\right)^2}\right).$$

### Symbols

| | | |
|---|---|---|
| $E$ | Electric field strength | $\frac{\mathrm{N}}{\mathrm{C}}$ |
| $E_{(+)}, E_{(-)}$ | Contribution to $E$ from the positive and negative part of the dipole | $\frac{\mathrm{N}}{\mathrm{C}}$ |
| $\epsilon_0$ | Permittivity constant | $8{,}85 \cdot 10^{-12}\,\mathrm{C}^2/\mathrm{N}\,\mathrm{m}^2$ |
| $z$ | Distance along axis | m |
| $d$ | Separation distance | m |
| $q$ | Charge | C |

### 2.3.3 Electric Field due to a Dipole along Dipole axis at Long Distance

$$E = \frac{1}{2\pi\epsilon_0}\frac{qd}{z^3} = \frac{1}{2\pi\epsilon_0}\frac{p}{z^3}.$$

### Assumptions

- $z \gg d$, i.e. the distance between charges is much smaller then the distance from the charges you are measuring ("normal" case)

### Symbols

| | | |
|---|---|---|
| $E$ | Electric field strength | $\frac{\mathrm{N}}{\mathrm{C}}$ |
| $\epsilon_0$ | Permittivity constant | $8{,}85 \cdot 10^{-12}\,\mathrm{C^2/N\,m^2}$ |
| $z$ | Distance along dipole axis | m |
| $d$ | Separation distance | m |
| $q$ | Charge | C |
| $p$ | Electric dipole moment ($p = qd$) | C m |

#### 2.3.4 Electric Field of a Charged Ring at Large Distance

$$E = \frac{1}{4\pi\epsilon_0}\frac{q}{z^2}.$$

## Assumptions

- $z \gg R$, i.e. distance from the ring is much grater than the radius of the ring

## Symbols

| | | |
|---|---|---|
| $E$ | Electric field strength | $\frac{\mathrm{N}}{\mathrm{C}}$ |
| $\epsilon_0$ | Permittivity constant | $8{,}85 \cdot 10^{-12}\,\mathrm{C^2/N\,m^2}$ |
| $q$ | Size of charge in the ring | C |
| $z$ | Distance perpendicular to the radius of the ring | m |

#### 2.3.5 Electric Field of a Charged Disk

$$E = \frac{\sigma}{2\epsilon_0}\left(1 - \frac{z}{\sqrt{z^2 + R^2}}\right).$$

## Assumptions

- Infinitesimally thin disk

## Symbols

| | | |
|---|---|---|
| $E$ | Electric field strength | $\frac{\mathrm{N}}{\mathrm{C}}$ |
| $\epsilon_0$ | Permittivity constant | $8{,}85 \cdot 10^{-12}\,\mathrm{C^2/N\,m^2}$ |
| $\sigma$ | Surface charge density | $\mathrm{C/m^2}$ |
| $z$ | Distance perpendicular to the disk | m |
| $R$ | radius of disk | m |

#### 2.3.6 Electric field from an infinite sheet

$$E = \frac{\sigma}{2\epsilon_0}.$$

## Assumptions

- Infinitesimally thin infinite sheet

## Symbols

| | | |
|---|---|---|
| $E$ | Electric field strength | $\frac{\mathrm{N}}{\mathrm{C}}$ |
| $\sigma$ | Surface charge density | $\mathrm{C/m^2}$ |
| $\epsilon_0$ | Permittivity constant | $8{,}85 \cdot 10^{-12}\,\mathrm{C^2/N\,m^2}$ |

#### 2.3.7 Electric field of a Conducting Sphere

$$E = \frac{1}{4\pi\epsilon_0}\frac{q}{r^2} = k\frac{q}{r^2}.$$

**Assumptions**

- Positive electric charge $Q$ distributed uniformly throughout the volume of a conducting sphere.
- Spherical symmetry
- No charge enclosed in conductor

**Symbols**

| | | |
|---|---|---|
| $E$ | Electric field strength | $\frac{\mathrm{N}}{\mathrm{C}}$ |
| $\epsilon_0$ | Permittivity constant | $8{,}85 \cdot 10^{-12}\,\mathrm{C}^2/\mathrm{N\,m}^2$ |
| $q$ | Charge | C |
| $r$ | Distance | m |
| $k$ | Coulomb constant | $8{,}99 \cdot 10^{9}\,\mathrm{N\,m}^2/\mathrm{C}^2$ |

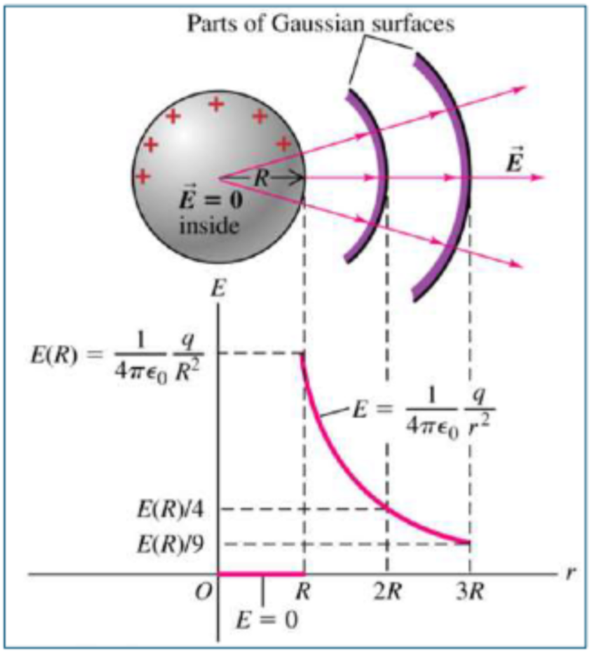


Figure 1: E-field propagation from a conducting sphere.

### 2.4 Electric Field of an Insulating Sphere

$$\text{Outside}: \quad E = \frac{1}{4\pi\epsilon_0}\frac{q}{r^2}$$
$$\text{Inside}: \quad E = \frac{q}{4\pi\epsilon_0 R^3} r.$$

**Assumptions**

- Positive electric charge $Q$ distributed uniformly throughout the volume of an insulating sphere.
- Spherical symmetry

**Symbols**

| | | |
|---|---|---|
| $E$ | Electric field strength | $\frac{\mathrm{N}}{\mathrm{C}}$ |
| $r$ | Distance | m |
| $R$ | Radius | m |
| $\epsilon_0$ | Permittivity constant | $8{,}85 \cdot 10^{-12}\,\mathrm{C^2/N\,m^2}$ |
| $q$ | Charge | C |

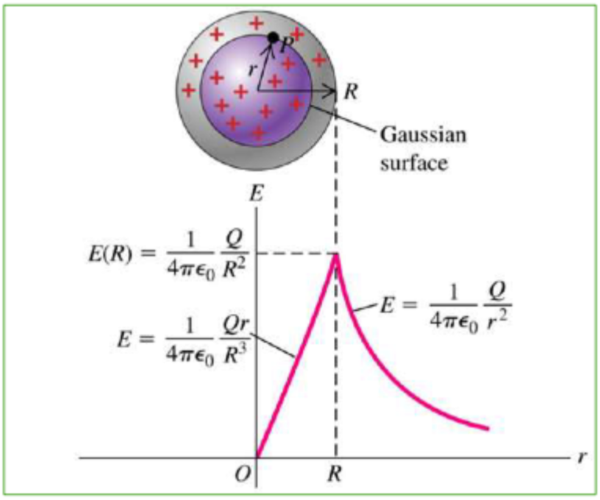


Figure 2: E-field propagation through an insulating sphere.

## 2.5 Gauss' Law

$$\Phi = \frac{q_{\mathrm{enc}}}{\epsilon_0}.$$

### Assumptions

- Vacuum (air)

### Symbols

| | | |
|---|---|---|
| $\Phi$ | Electric flux | V m |
| $q_{\mathrm{enc}}$ | Enclosed charge | C |
| $\epsilon_0$ | Permittivity constant | $8{,}85 \cdot 10^{-12}\,\mathrm{C^2/N\,m^2}$ |

### 2.5.1 Net electric flux through a surface caused by an electric field

$$\Phi = \oint \boldsymbol{E} \cdot \mathrm{d}\boldsymbol{A}.$$

### Symbols

| | | |
|---|---|---|
| $\Phi$ | Electric flux | V m |
| $\boldsymbol{E}$ | Electric field vector | $\frac{\mathrm{N}}{\mathrm{C}}$ |
| $\boldsymbol{A}$ | Area vector | $\mathrm{m^2}$ |

## 2.6 Electric Potential Energy

$$U_E = q_0 E \Delta d.$$

#### Symbols

| | | |
|---|---|---|
| $U_E$ | Electric potential energy | J |
| $q_0$ | Test charge | C |
| $E$ | Electric field strength | $\frac{\mathrm{N}}{\mathrm{C}}$ |
| $d$ | Separation distance | m |

### 2.6.1 Electrostatic Potential

$$V_E = \frac{U_E}{q_0}.$$

#### Symbols

| | | |
|---|---|---|
| $V_E$ | Electrostatic potential | V |
| $U_E$ | Electric potential energy | J |
| $q_0$ | Test charge | C |

### 2.6.2 Electrostatic Potential of a Plate Capacitor

$$V_E = Ed.$$

#### Symbols

| | | |
|---|---|---|
| $V_E$ | Electrostatic potential | V |
| $E$ | Electric field strength | $\frac{\mathrm{N}}{\mathrm{C}}$ |
| $d$ | Separation distance | m |

# Lecture 2: Fundamentals of Electric Circuits

28. Januar 2026

## 3.1 Current, Voltage, and Resistance

### 3.1.1 Electric Current

$$I = \frac{\mathrm{d}q}{\mathrm{d}t}.$$

#### Symbols

| | | |
|---|---|---|
| $I$ | Current | A |
| $q$ | Charge | C |
| $t$ | Time | s |

### 3.1.2 Electric Voltage

$$V = \frac{\mathrm{d}W}{\mathrm{d}Q}.$$

#### Symbols

| | | |
|---|---|---|
| $V$ | Voltage | V |
| $W$ | Work | J |
| $Q$ | Total charge | C |

### 3.1.3 Electric Resistance

$$R = \frac{V}{I}.$$

#### Symbols

| | | |
|---|---|---|
| $R$ | Resistance | $\Omega$ |
| $V$ | Voltage | V |
| $I$ | Current | A |

### 3.1.4 Current Density

$$J = \frac{I}{A}.$$

#### Symbols

| | | |
|---|---|---|
| $J$ | Current density | $\mathrm{A/m^2}$ |
| $I$ | Current | A |
| $A$ | Cross-sectional area | $\mathrm{m^2}$ |

### 3.1.5 Resistivity

$$\rho = \frac{E}{J}.$$

#### Symbols

| | | |
|---|---|---|
| $\rho$ | Resistivity | $\Omega\,\mathrm{m}$ |
| $E$ | Electric field strength | $\frac{\mathrm{N}}{\mathrm{C}}$ |
| $J$ | Current density | $\mathrm{A/m^2}$ |

### 3.1.6 Conductivity

$$\sigma = \frac{1}{\rho}.$$

#### Symbols

| | | |
|---|---|---|
| $\sigma$ | Conductivity | $\Omega^{-1}\,\mathrm{m}^{-1}$ |
| $\rho$ | Resistivity | $\Omega\,\mathrm{m}$ |

### 3.1.7 Resistance From Resistivity

$$R = \frac{\rho L}{A}.$$

#### Symbols

| | | |
|---|---|---|
| $R$ | Resistance | $\Omega$ |
| $\rho$ | Resistivity | $\Omega\,\mathrm{m}$ |
| $L$ | Length | m |
| $A$ | Cross-sectional area | $\mathrm{m^2}$ |

#### 3.1.8 Ohm's Law

$$I \propto V.$$

**Symbols**

| | | |
|---|---|---|
| $I$ | Current | A |
| $V$ | Voltage | V |

#### 3.1.9 Electric Power

$$P = VI = I^2 R = \frac{V^2}{R}.$$

**Symbols**

| | | |
|---|---|---|
| $P$ | Electrical power | W |
| $V$ | Voltage | V |
| $I$ | Current | A |
| $R$ | Resistance | Ω |

### 3.2 Kirchoff's Circuit Laws

#### 3.2.1 Kirchoff's Current Law (KCL)

$$\sum_{n=1}^{N} I_n = 0.$$

**Symbols**

| | | |
|---|---|---|
| $I_n$ | Current at node $n$ | A |
| $N$ | Amount of nodes | — |

#### 3.2.2 Kirchoff's Voltage Law (KVL)

$$\sum_{n=1}^{N} V_n = 0.$$

**Symbols**

| | | |
|---|---|---|
| $V_n$ | Voltage difference between two points | V |
| $N$ | Amount of points | N |

#### 3.2.3 Resistors in Series

$$R_{\text{EQ}} = \sum_{n=1}^{N} R_n.$$

**Symbols**

| | | |
|---|---|---|
| $R_{\text{eq}}$ | Equivalent resistance | Ω |
| $R_n$ | Resistance of resistor $n$ | Ω |
| $N$ | Amount of resistors in series | — |

#### 3.2.4 Voltage Divider Rule

$$V_n = \frac{R_n}{\sum_{n=1}^{N} R_n} V_s.$$

**Symbols**

| | | |
|---|---|---|
| $V_n$ | Voltage difference over resistor $n$ | V |
| $R_n$ | Resistance over resistor $n$ | Ω |
| $V_s$ | Source voltage | V |

#### 3.2.5 Resistors in Parallel

$$R_{\mathrm{EQ}} = \frac{1}{\sum_{n=1}^{N} \frac{1}{R_n}}.$$

**Symbols**

| | | |
|---|---|---|
| $R_{\mathrm{eq}}$ | Equivalent resistance | Ω |
| $R_n$ | Resistance of resistor $n$ | Ω |
| $N$ | Amount of resistors in parallel | — |

#### 3.2.6 Current Divider Rule

$$I_n = \frac{\frac{1}{R_n}}{\sum_{n=1}^{N} \frac{1}{R_n}} I_s.$$

**Symbols**

| | | |
|---|---|---|
| $I_n$ | Current difference over resistor $n$ | A |
| $R_n$ | Resistance over resistor $n$ | Ω |
| $I_s$ | Source current | A |

## Lecture 3: Principles of Circuit Analysis — 2. Februar 2026

## Lecture 4: Capacitors & Inductors — 9. Februar 2026

### 5.1 Capacitors

#### 5.1.1 Capacitance

$$C = \frac{q}{V}.$$

**Symbols**

| | | |
|---|---|---|
| $C$ | Capacitance | F |
| $q$ | Charge | C |
| $V$ | Voltage | V |

#### 5.1.2 Capacitance of a Parallel-Plate Capacitor

$$C = \epsilon_0 \frac{A}{d}.$$

### Symbols

| | | |
|---|---|---|
| $C$ | Capacitance | F |
| $\epsilon_0$ | Permittivity constant | $8{,}85 \cdot 10^{-12}\,\mathrm{C^2/N\,m^2}$ |
| $A$ | Cross-sectional area | $\mathrm{m^2}$ |
| $d$ | Separation distance | m |

### 5.1.3 I-V Relation for an Ideal Capacitor

$$I_C(t) = C\frac{\mathrm{d}V_C(t)}{\mathrm{d}t} \iff V_C(t) = \frac{1}{C}\int_{-\infty}^{t} I_C(t')\,\mathrm{d}t'.$$

### Symbols

| | | |
|---|---|---|
| $I_C$ | Current through capacitor | A |
| $V_C$ | Voltage over capacitor | V |
| $C$ | Capacitance | F |
| $t$ | Time | s |

### 5.1.4 Capacitance of a Cylindrical Capacitor

$$C = 2\pi\epsilon_0 \frac{L}{\ln\frac{b}{a}}.$$

### Symbols

| | | |
|---|---|---|
| $C$ | Capacitance | F |
| $\epsilon_0$ | Permittivity constant | $8{,}85 \cdot 10^{-12}\,\mathrm{C^2/N\,m^2}$ |
| $L$ | Length | m |
| $a$ | Radius of the inner post | m |
| $b$ | Radius of the outer post | m |

### 5.1.5 Capacitance of a Cylindrical Capacitor

$$C = 4\pi\epsilon_0 \frac{ab}{b-a}.$$

### Symbols

| | | |
|---|---|---|
| $C$ | Capacitance | F |
| $\epsilon_0$ | Permittivity constant | $8{,}85 \cdot 10^{-12}\,\mathrm{C^2/N\,m^2}$ |
| $a$ | Radius of the inner sphere | m |
| $b$ | Radius of the outer sphere | m |

### 5.1.6 Parallel Capacitance

$$C_{\mathrm{eq}} = \sum_{i=1}^{n} C_i.$$

### Symbols

| | | |
|---|---|---|
| $n$ | Number of capacitors | — |
| $C_{eq}$ | Equivalent capacitance | F |
| $C_i$ | Capacitance of capacitor $i$ | F |

### 5.1.7 Capacitors in Series

$$\frac{1}{C_{\text{eq}}} = \sum_{i=1}^{n} \frac{1}{C_i}.$$

#### Symbols

| | | |
|---|---|---|
| $n$ | Number of capacitors | — |
| $C_{eq}$ | Equivalent capacitance | F |
| $C_i$ | Capacitance of capacitor $i$ | F |

### 5.1.8 Potential Energy in a Capacitor

$$U_E = \frac{1}{2}CV^2.$$

#### Symbols

| | | |
|---|---|---|
| $U_E$ | Electric potential energy | J |
| $C$ | Capacitance | F |
| $V$ | Voltage | V |

## 5.2 Inductors

### 5.2.1 Inductance

$$L = n\frac{\Phi_B}{I}.$$

#### Symbols

| | | |
|---|---|---|
| $L$ | Inductance | H |
| $n$ | Number of windings | — |
| $\Phi_B$ | Magnetic flux | T |
| $I$ | Current | A |

### 5.2.2 I-V Relation for an Ideal Inductor

$$V_L(t) = L\frac{\mathrm{d}I_L(t)}{\mathrm{d}t} \iff I_L(t) = \frac{1}{L}\int_{t_0}^{t} V_L(t')\,\mathrm{d}t' + I_0.$$

#### Symbols

| | | |
|---|---|---|
| $I_L$ | Current through inductor | A |
| $V_L$ | Voltage over inductor | V |
| $L$ | Inductance | H |
| $t$ | Time | s |

### 5.2.3 Magnetic Field of an Inductor

$$\boldsymbol{B} = \mu_0 I n.$$

#### Symbols

| | | |
|---|---|---|
| $\boldsymbol{B}$ | Magnetic field vector | T |

| | | |
|---|---|---|
| $\mu_0$ | Magnetic vacuum permeability | $1{,}26 \cdot 10^{-6}\,\mathrm{N/A^2}$ |
| $I$ | Current | A |
| $n$ | Number of windings | — |

### 5.2.4 Inductors in Series

$$L_{\mathrm{eq}} = \sum_{i=1}^{n} L_i.$$

## Symbols

| | | |
|---|---|---|
| $n$ | Number of inductors | — |
| $L_{eq}$ | Equivalent Inductance | H |
| $L_i$ | Inductance of inductor $i$ | H |

### 5.2.5 Parallel Inductors

$$\frac{1}{L_{\mathrm{eq}}} = \sum_{i=1}^{n} \frac{1}{L_i}.$$

## Symbols

| | | |
|---|---|---|
| $n$ | Number of inductors | — |
| $L_{eq}$ | Equivalent Inductance | H |
| $L_i$ | Inductance of inductor $i$ | H |

### 5.2.6 Energy Stored in an Inductor

$$U_L = \frac{1}{2} L I^2.$$

## Symbols

| | | |
|---|---|---|
| $U_L$ | Potential energy stored in inductor | J |
| $L$ | Inductance | H |
| $I$ | Current | A |

### 5.2.7 Energy Density in an Inductor

$$u_L = \frac{\boldsymbol{B}}{2\mu_0}.$$

## Symbols

| | | |
|---|---|---|
| $u_L$ | Energy density of inductor | $\mathrm{J/m^3}$ |
| $\boldsymbol{B}$ | Magnetic field vector | T |
| $\mu_0$ | Magnetic vacuum permeability | $1{,}26 \cdot 10^{-6}\,\mathrm{N/A^2}$ |

## 5.3 RC and RL Circuits in the DC Regime

### 5.3.1 Charge in a (Charging) Capacitor as a Function of Time

$$q(t) = CV_s\left(1 - e^{-\frac{t}{RC}}\right).$$

### Symbols

| | | |
|---|---|---|
| $q$ | Charge | C |
| $C$ | Capacitance | F |
| $V_s$ | Source voltage | V |
| $t$ | Time | s |
| $R$ | Resistance | Ω |
| $C$ | Capacitance | F |

#### 5.3.2 Current in a (Charging) Capacitor as a Function of Time

$$I_C(t) = \frac{V_s}{R} e^{-\frac{t}{RC}}.$$

### Symbols

| | | |
|---|---|---|
| $I_C$ | Current through capacitor | A |
| $C$ | Capacitance | F |
| $V_s$ | Source voltage | V |
| $t$ | Time | s |
| $R$ | Resistance | Ω |
| $C$ | Capacitance | F |

#### 5.3.3 Capacitive Time Constant

$$\tau_C = RC.$$

### Symbols

| | | |
|---|---|---|
| $\tau_C$ | Capacitive Time Constant | s |
| $R$ | Resistance | Ω |
| $C$ | Capacitance | F |

#### 5.3.4 Inductive Time Constant

$$\tau_L = \frac{L}{R}.$$

### Symbols

| | | |
|---|---|---|
| $\tau_L$ | Inductive Time Constant | s |
| $L$ | Inductance | H |
| $R$ | Resistance | Ω |

#### 5.3.5 Current in an Inductor

$$I = \frac{V_s}{R}\left(1 - e^{-\frac{t}{\tau_L}}\right).$$

### Symbols

| | | |
|---|---|---|
| $I$ | Current | A |
| $V_s$ | Source voltage | V |
| $R$ | Resistance | Ω |

| | | |
|---|---|---|
| $t$ | Time | s |
| $\tau_L$ | Inductive Time Constant | s |

# Lecture 5: AC Circuits

11. Februar 2026

## 6.1 AC Circuits

### 6.1.1 Periodic Signals

$$x(t) = x(t + nT), \quad n = 1, 2, 3, \dots.$$

**Symbols**

| | | |
|---|---|---|
| $x(t)$ | Periodic signal $x$ of $t$ | — |
| $t$ | Time | s |
| $T$ | Period | s |

### 6.1.2 Average Value of a Signal Waveform

$$\langle x(t) \rangle = \frac{1}{T} \int_0^T x(t')\,\mathrm{d}t'.$$

**Symbols**

| | | |
|---|---|---|
| $x(t)$ | Periodic signal $x$ of $t$ | — |
| $\langle x(t) \rangle$ | Average value of $x(t)$ | — |
| $T$ | Period | s |

### 6.1.3 RMS value

$$x_{\text{rms}} = \sqrt{\frac{1}{T} \int_0^T x^2(t')\,\mathrm{d}t'}.$$

**Symbols**

| | | |
|---|---|---|
| $x_{\text{rms}}$ | RMS of $x(t)$ | — |
| $x(t)$ | Periodic signal $x$ of $t$ | — |
| $T$ | Period | s |

## 6.2 Impedance

### 6.2.1 Impedance of a Resistor

$$Z(j\omega) = \frac{\boldsymbol{V}(j\omega)}{\boldsymbol{I}(j\omega)} = R.$$

**Symbols**

| | | |
|---|---|---|
| $Z$ | Impedance | $\Omega$ |
| $j$ | Imaginary unit | — |
| $\omega$ | Frequency | Hz |
| $V$ | Voltage | V |
| $I$ | Current | A |

| | | |
|---|---|---|
| $R$ | Resistance | $\Omega$ |

### 6.2.2 Impedance of an Inductor

$$Z(j\omega) = \frac{\boldsymbol{V}(j\omega)}{\boldsymbol{I}(j\omega)} = \omega L \angle \frac{\pi}{2}.$$

#### Symbols

| | | |
|---|---|---|
| $Z$ | Impedance | $\Omega$ |
| $j$ | Imaginary unit | — |
| $\omega$ | Frequency | Hz |
| $V$ | Voltage | V |
| $I$ | Current | A |
| $L$ | Inductance | H |

### 6.2.3 Impedance of a Capacitor

$$Z(j\omega) = \frac{\boldsymbol{V}(j\omega)}{\boldsymbol{I}(j\omega)} = \frac{1}{\omega C} \angle - \frac{\pi}{2} = \frac{1}{j\omega C}.$$

#### Symbols

| | | |
|---|---|---|
| $Z$ | Impedance | $\Omega$ |
| $j$ | Imaginary unit | — |
| $\omega$ | Frequency | Hz |
| $V$ | Voltage | V |
| $I$ | Current | A |
| $C$ | Capacitance | F |

### 6.2.4 Total Impedance

$$Z(j\omega) = R(j\omega) + jX(j\omega).$$

#### Symbols

| | | |
|---|---|---|
| $Z$ | Impedance | $\Omega$ |
| $j$ | Imaginary unit | — |
| $\omega$ | Frequency | Hz |
| $R$ | Resistance | $\Omega$ |
| $X$ | Reactance | $\Omega$ |

### 6.2.5 Admittance

$$Y = \frac{1}{Z}.$$

#### Symbols

| | | |
|---|---|---|
| $Y$ | Admittance | § |
| $Z$ | Impedance | $\Omega$ |

## 6.3 Transient Analysis

### 6.3.1 1st Order Differential Equation for RC Circuits

$$\tau \frac{\mathrm{d}x(t)}{\mathrm{d}t} + x(t) = K_s f(t).$$

#### Symbols

| | | |
|---|---|---|
| $\tau$ | $\tau = a_1 / a_0$ is a time constant | — |
| $x$ | Voltage or current | — |
| $t$ | Time | s |
| $K_s$ | $K_s = b_0 / a_0$ is DC gain | — |

### 6.3.2 2nd Order Differential Equation for RLC Circuits

$$\frac{1}{\omega_n^2} \frac{\mathrm{d}x^2(t)}{\mathrm{d}t^2} + \frac{2\zeta}{\omega_n} \frac{\mathrm{d}x(t)}{\mathrm{d}t} = K_s f(t).$$

#### Symbols

| | | |
|---|---|---|
| $\omega_n$ | $\omega_n = \sqrt{a_0 / a_2}$ is the natural frequency | Hz |
| $\zeta$ | $\zeta = a_1 / 2\sqrt{1/(a_0 a_2)}$ is the damping ratio | — |
| $x$ | Voltage or current | — |
| $t$ | Time | s |
| $K_s$ | $K_s = b_0 / a_0$ is DC gain | — |